

MATHEMATICAL ANALYSIS 1
HOMEWORK 1

(1) Prove that there are infinitely many prime numbers (Euclid's Theorem).

(2) Prove the following lemma by contradiction:

Lemma 0.1. *Let $I \subseteq \mathbb{R}$ be a subset of $\mathbb{R}$ that satisfies the following condition: for any $x_1, x_2 \in I$ with $x_1 < x_2$, the entire closed interval $[x_1, x_2]$ is a subset of I . Then I is an interval, i.e. I could be an open, closed, or half-open-half-closed interval, and one or both of its endpoints can be infinite.*

(3) Prove that the solution of the equation $x^2 = 7$ is irrational.

(4) Prove that for any set X and for any two subsets $A, B \subset X$, we have $(A \cup B)^C = A^C \cap B^C$.

(5) **Quantifiers.**

(a) Write the negation of the statement

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N}_+ \quad \text{such that} \quad \frac{1}{N} < \varepsilon,$$

moving the negation all the way inside (i.e. your answer should not contain the symbol $\neg$).

Which of the two statements (the original or its negation) is true? Explain.

(b) Determine whether each of the following statements is true or false, and explain the difference between them:

(i) $\forall x \in \mathbb{R} \quad \exists y \in \mathbb{R} \quad \text{such that} \quad x < y,$ (ii) $\exists y \in \mathbb{R} \quad \text{such that} \quad \forall x \in \mathbb{R}, \quad x < y.$

(6) Prove by induction that $2^n \geq n^2$ for all $n \in \mathbb{N}$ with $n \geq 4$. Is the inequality true for $n = 3$?

(7) Solve the inequality: $\sqrt{|x^2 - 4|} - x \geq 0$.

(8) Let $A \subseteq \mathbb{R}$ be nonempty and bounded from above, and suppose that $s_1, s_2 \in \mathbb{R}$ both satisfy the two properties in the definition of the supremum of A . Prove that $s_1 = s_2$. (This is the uniqueness part of Proposition 2.8 in the lecture notes.)

Hint: prove by contradiction.

(9) Let $P \subseteq \mathbb{R}$ be the set of prime numbers.

(a) Is P bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to P ?

(b) Let $Q = \{x \in \mathbb{R} \mid x^{-1} \in P\}$. Is Q bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to Q ?

(10) Consider the following subset of $\mathbb{R}$:

$$A = \left\{ x \in \mathbb{R} \mid 0 \leq x < 1 \quad \text{or} \quad x = \frac{2n-3}{n-1}, \quad n \in \mathbb{N} \setminus \{0, 1\} \right\}.$$

Is A bounded from above? from below? what is its infimum? supremum? do either the supremum or infimum belong to A ?

SOLUTIONS

- (1) We prove by contradiction. Suppose there are finitely many primes $p_1, p_2, \dots, p_n$. Let $N = p_1 p_2 \cdots p_n + 1$. Then $N > 1$ and N is not divisible by any p_i (since division by p_i leaves remainder 1). Thus either N is prime or has a prime divisor different from all p_i , contradicting the assumption that $p_1, \dots, p_n$ are all primes.
- (2) If $I = \emptyset$ there is nothing to prove (the empty set is regarded as a degenerate interval), so assume $I \neq \emptyset$. Let $a = \inf I$ and $b = \sup I$. By the Completeness Axiom, $a \in \mathbb{R}$ if I is bounded from below and $a = -\infty$ otherwise; similarly $b \in \mathbb{R}$ if I is bounded from above and $b = +\infty$ otherwise. If $a = b$, then $a, b \in \mathbb{R}$ and every $x \in I$ satisfies $a \leq x \leq b = a$, so $I = \{a\} = [a, a]$. From now on assume $a < b$.

Step 1: every $z \in \mathbb{R}$ with $a < z < b$ belongs to I . Suppose, by contradiction, that there is $z \in \mathbb{R}$ with $a < z < b$ and $z \notin I$. Since $z > a$, z is not a lower bound of I : if $a \in \mathbb{R}$ this is property 2 of the infimum, and if $a = -\infty$ it is because I is not bounded from below. Hence there exists $x_1 \in I$ with $x_1 < z$. Similarly, since $z < b$, z is not an upper bound of I , so there exists $x_2 \in I$ with $x_2 > z$. Now $x_1, x_2 \in I$ with $x_1 < x_2$, so by assumption $[x_1, x_2] \subseteq I$. But $x_1 < z < x_2$, so $z \in [x_1, x_2] \subseteq I$, a contradiction.

Step 2: conclusion. If $a \in \mathbb{R}$, then a is a lower bound of I , so $x \geq a$ for all $x \in I$; if $b \in \mathbb{R}$, then $x \leq b$ for all $x \in I$. Combining this with Step 1, we distinguish four cases:

- $a, b \in \mathbb{R}$: then $(a, b) \subseteq I \subseteq [a, b]$, so I is one of (a, b) , $[a, b)$, $(a, b]$, $[a, b]$, according to whether $a \in I$ and whether $b \in I$.
- $a \in \mathbb{R}, b = +\infty$: then $(a, +\infty) \subseteq I \subseteq [a, +\infty)$, so $I = (a, +\infty)$ or $I = [a, +\infty)$.
- $a = -\infty, b \in \mathbb{R}$: then $(-\infty, b) \subseteq I \subseteq (-\infty, b]$, so $I = (-\infty, b)$ or $I = (-\infty, b]$.
- $a = -\infty, b = +\infty$: then every real number belongs to I , so $I = \mathbb{R} = (-\infty, +\infty)$.

In all cases I is an interval.

- (3) We prove by contradiction. Suppose $x^2 = 7$ has a rational solution $x = \frac{p}{q}$ where $p, q \in \mathbb{N}$ have no common divisors. Then $p^2 = 7q^2$, so $7 \mid p^2$. Since 7 is prime, $7 \mid p$, say $p = 7k$. Then $49k^2 = 7q^2$, so $7k^2 = q^2$ and $7 \mid q^2$, hence $7 \mid q$. This contradicts the assumption that p and q have no common divisors.
- (4) As in the proof we did in class, to show equality of two sets we show that they both contain one another. Let $x \in (A \cup B)^C$. Then $x \notin A \cup B$, so $x \notin A$ and $x \notin B$. Thus $x \in A^C$ and $x \in B^C$, so $x \in A^C \cap B^C$. Conversely, if $x \in A^C \cap B^C$, then $x \notin A$ and $x \notin B$, so $x \notin A \cup B$, hence $x \in (A \cup B)^C$.
- (5) (a) The negation is

$$\exists \varepsilon > 0 \text{ such that } \forall N \in \mathbb{N}_+, \frac{1}{N} \geq \varepsilon.$$

The original statement is true (so the negation is false): given $\varepsilon > 0$, let N be any natural number with $N > \frac{1}{\varepsilon}$ (e.g. $N = \lceil \frac{1}{\varepsilon} \rceil + 1$). Then $\frac{1}{N} < \varepsilon$. This is the same idea as in Lemma 2.7 of the lecture notes.

- (b) (i) is true: given $x \in \mathbb{R}$, take $y = x + 1$. (ii) is false: its negation is “ $\forall y \in \mathbb{R} \exists x \in \mathbb{R}$ such that $x \geq y$ ”, which is true (take $x = y$). The difference is the order of the quantifiers: in (i) y is allowed to depend on x , while in (ii) a single y would have to be larger than every real number, i.e. $\mathbb{R}$ would have to be bounded from above.

- (6) Let $p(n) = '2^n \geq n^2'$.

Base case $n = 4$: $2^4 = 16 \geq 16 = 4^2$, so $p(4)$ is true.

Induction step. Let $n \geq 4$ and assume $p(n)$, i.e. $2^n \geq n^2$. Then

$$2^{n+1} = 2 \cdot 2^n \geq 2n^2.$$

It remains to show that $2n^2 \geq (n+1)^2$, i.e. $n^2 - 2n - 1 \geq 0$, i.e. $(n-1)^2 \geq 2$. This holds since $n \geq 4$ gives $(n-1)^2 \geq 9$. Hence $2^{n+1} \geq (n+1)^2$, so $p(n+1)$ is true. By induction, $2^n \geq n^2$ for all $n \geq 4$.

For $n = 3$ the inequality is false: $2^3 = 8 < 9 = 3^2$. (It does hold for $n = 0, 1, 2$, which shows that the base case matters: the induction step itself fails at $n = 2$, where $(n-1)^2 = 1 < 2$.)

- (7) The inequality is $\sqrt{|x^2 - 4|} \geq x$. For $x \leq 0$, this holds automatically. For $x > 0$, we require $|x^2 - 4| \geq x^2$, which gives $x^2 - 4 \leq -x^2$ or $x^2 - 4 \geq x^2$. The latter is impossible. The former gives $2x^2 \leq 4$, so $x^2 \leq 2$, i.e., $0 < x \leq \sqrt{2}$. Therefore the solution is $x \in (-\infty, \sqrt{2}]$.

- (8) Suppose, by contradiction, that $s_1 \neq s_2$; without loss of generality $s_1 < s_2$. Apply property 2 of the definition to s_2 with $r = s_1 < s_2$: there exists $x \in A$ with $x > s_1$. But property 1 for s_1 says that $x \leq s_1$ for all $x \in A$, a contradiction. Hence $s_1 = s_2$. (The proof of uniqueness of the infimum is analogous.)
- (9) (a) P is not bounded from above (there are infinitely many primes). P is bounded from below by 0 (or by 2). We have $\inf P = 2 \in P$, and $\sup P$ does not exist in $\mathbb{R}$ (or $\sup P = +\infty$).
- (b) $Q = \{\frac{1}{p} \mid p \in P\}$ is bounded from above by $\frac{1}{2}$ and from below by 0. We have $\sup Q = \frac{1}{2} \in Q$ and $\inf Q = 0 \notin Q$ (since Q consists of reciprocals of primes, $0 \notin Q$).
- (10) Note that $\frac{2n-3}{n-1} = \frac{2(n-1)-1}{n-1} = 2 - \frac{1}{n-1}$. For $n \geq 2$, this sequence increases toward 2 from below. Thus $A = [0, 1) \cup \{1, \frac{3}{2}, \frac{5}{3}, \frac{7}{4}, \dots\}$ where the sequence approaches 2. We have $\inf A = 0 \in A$, and $\sup A = 2 \notin A$ (since all values $\frac{2n-3}{n-1} < 2$ for $n \geq 2$). A is bounded both from above and from below.